

CHIRANTH D NANDI

B. TECH STUDENT, CSE(AIML)

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SUMMARY

Second year B.Tech CSE(AIML) student with experience in building AI and full stack applications. Proficient in Python and C, with coursework in data structures and MERN stack. Worked with machine learning, vector databases, computer vision, and cybersecurity. Interested in applying skills to cybersecurity problems and open source contribution. Seeking AIML internships or Software Engineering Internship opportunities.

EDUCATION

2024-2028 PES University, Bengaluru

CGPA: 7.21

2024 Class 12 PU Board Examination, KSEAB

Percentage: 97%

2022 Class 10 Board Examination, CBSE

Percentage: 96.8%

EXPERIENCE

SDE Intern (Onsite), Crescenzia Wellness Private Limited | December 2025 – Present

- Developed an end to end order to fulfillment automation pipeline for B2B clients, reducing manual workflows by 70% while implementing secure 2FA and role based access controls. Working with MERN stack, MySQL, JWT Auth, RESTful APIs, Document parsing from PDFs, Regular Expression based validation.

Technical Team Member, Neural Hive PESU | November 2025 – Present

- Conducted technical workshops and hackathons for 50+ students and collaborated with a 11 member team on AI projects and research.

SKILLS

- Languages:** Python, C Programming, JavaScript, MySQL, HTML5.
- Libraries and Frameworks:** PyTorch, React, Django, Node.js, Express.js, MongoDB, LangChain, OpenCV, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, GTK3.
- AIML Technologies:** Machine Learning, Computer Vision, Retrieval Augmented Generation (RAG), Vector Databases, Model Quantization. Generative AI, Multimodal LLM.
- Tools:** Model Context Protocol (MCP), AES Encryption, OpenSSL, Haar Cascade, Git.
- Interests:** Healthcare AI, Cybersecurity, Opensource

PROJECTS

Automated AI Model Red-Teaming and Defense Validation System. [GitHub Repo](#)

- Built an LLM security scanner running supply chain artifact analysis, AST-based code inspection, OSV.dev CVE lookups, and 35+ adversarial red-team probes with adaptive escalation and a 6-judge consensus panel, achieving 93% detection rate across seeded vulnerability benchmarks.
- Validated 5 mitigations against every successful attack and synthesized before/after AI-CVSS risk scores with OWASP LLM Top 10, MITRE ATLAS, and CWE compliance mapping into exportable forensic reports.
- Technologies:** Python, Streamlit, HuggingFace, Ollama, Groq, OWASP-10 GEN AI.

Autonomous Driving Software for Obstacle Avoidance using Reinforcement Learning [GitHub Repo](#)

- Developed a full autonomous driving pipeline with a custom Gymnasium environment simulating physics-based ultrasonic raycasting and relative velocity tracking to distinguish static obstacles from oncoming vehicles across a 3-lane scenario.
- Trained a Transformer + LSTM + RecurrentPPO policy via behavioral cloning pretraining on 98,000 expert demonstrations followed by 2M PPO steps, then exported to INT8 quantized ONNX Runtime for Raspberry Pi 4 deployment alongside a MobileNetV3-Small traffic sign classifier.
- Technologies:** Python, PyTorch, Stable-Baselines3, Gymnasium, ONNX Runtime, MobileNetV3, Raspberry Pi 4.

Data Vault: File Encryption using AES and Role Based Access System RBAC. [GitHub Repo](#)

- Developed desktop application with RBAC supporting admin and staff roles, implementing AES-256 encryption via OpenSSL with automated key management (256 bit keys, 128 bit IVs) and secure original file deletion.
- Built GUI in C using GTK3 with admin dashboards for decryption and audit logging of CSV data, packaged and distributed for Windows using Inno Setup, with zero external dependencies.
- Technologies:** C, GTK3, OpenSSL, AES-256 Encryption, File I/O, CSV Logging, Inno Setup.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** issued Oct'25 [view credential](#)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** issued Oct'25 [view credential](#)
- Fine Tuning LLMs for Production by Actueloop**
- Practical Web Security and Ethical Hacking by PESU I/O** issued Nov'25 [view credential](#)

RESEARCH AND PUBLICATIONS

Synthesis of Ag-PPy/TiO₂ Nanocomposite Polymers Employing Ocimum tenuiflorum Extract-Derived TiO₂: AC Conductivity, Humidity Sensor Application, & EMI Shielding Efficiency. (April 2026) [view credential](#)

- Synthesised Ag-PPy/TiO₂ Nanocomposite Polymer via Green Synthesis, and presented the paper at a **National Conference**, NCQTAM-26 conducted by the Karnataka Physics Association. Included structural characterization, AC conductivity, humidity sensor application, & EMI shielding properties. Experiments done under Dr. Revanasidappa at PES University, Electronic City Campus.

COMMUNITY INVOLVEMENT

- Published Aviation Photographer with **publications with Air India Express** (Tata Group), **Star Air** (Sanjay Ghodawat Group), and HAL Chief Test Pilot, and with community interactions across **Europe**, fostering **cross cultural collaboration**.